IE2044 - System Modeling and Prototyping

Final Assigmentgnment

Development of a Digital Thermometer Design (0°C – 99°C)

Group 11

IT24101644 - N.K.B.A.G.A.I.NARASINGHE

IT24103484 - B.Amsawardan

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1. Introduction

The project aims to develop a 2-digit digital thermometer system which can measure temperatures between 0°C and 99°C. The temperature measurement system uses LM35 temperature sensors together with ICL7107 analog-to-digital converters to process both analog and digital signals. The main objective was not only to simulate the system but also to convert it into a real PCB-based working circuit, ensuring practical understanding of signal behavior, noise handling, and hardware constraints. The project required careful design through calibration and grounding techniques and physical layout procedures which made it operate more like an actual engineering project than an academic study.

2. Objectives

- Design a temperature measurement system using LM35
- Convert analog voltage into readable digital output
- Calibrate system using reference voltage (Vref)
- Simulate the circuit in Proteus
- Design PCB using EasyEDA
- Ensure stable and noise-free output

3. Theoretical Background

3.1 LM35 Temperature Sensor

The LM35 is a precision temperature sensor that outputs a voltage linearly proportional to temperature.

- Scale factor: **10 mV per °C**
- Example:

  25°C → 250 mV

  50°C → 500 mV

This linear relationship simplifies calculations and removes the need for complex conversions.

3.2 ICL7107 ADC and Display Driver

The ICL7107 is a **dual-slope integrating ADC** that directly drives 7-segment displays.

Key advantages:

- High accuracy
- Noise rejection
- Direct display output (no microcontroller needed)

3.3 Dual-Slope Integration

The conversion works in two steps:

1. **Integration Phase**
   Input voltage charges a capacitor over fixed time
2. **De-integration Phase**
   Capacitor discharges using reference voltage

The time taken is proportional to input voltage → converted into digits.

This method is very stable and reduces noise errors.

4. Design Methodology

4.1 Voltage Scaling & Calibration

Since LM35 outputs **10 mV per °C**, we need:

10 mV → display "1"

100 mV → display "10"

So we set:

Reference Voltage (Vref) = 100 mV

This ensures:

Temperature (°C) = Display Value

4.2 Proteus Simulation

We designed the full circuit in Proteus including:

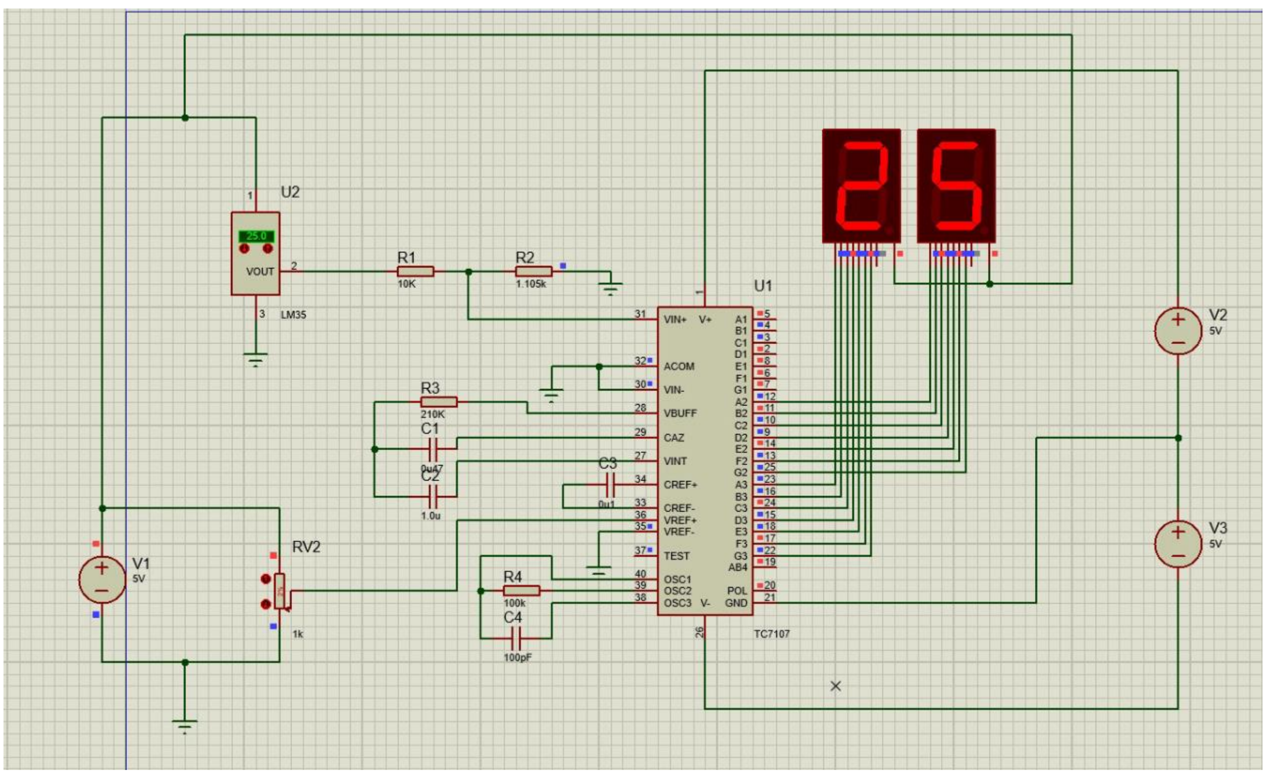
- LM35 sensor
- ICL7107 IC
- 2 x 7-segment displays
- Potentiometer for Vref

Testing was done by varying temperature input.

Results observed:

- 0°C → Display "00"
- 25°C → Display "25"
- 50°C → Display "50"
- 99°C → Display "99"

This confirmed correct voltage scaling.



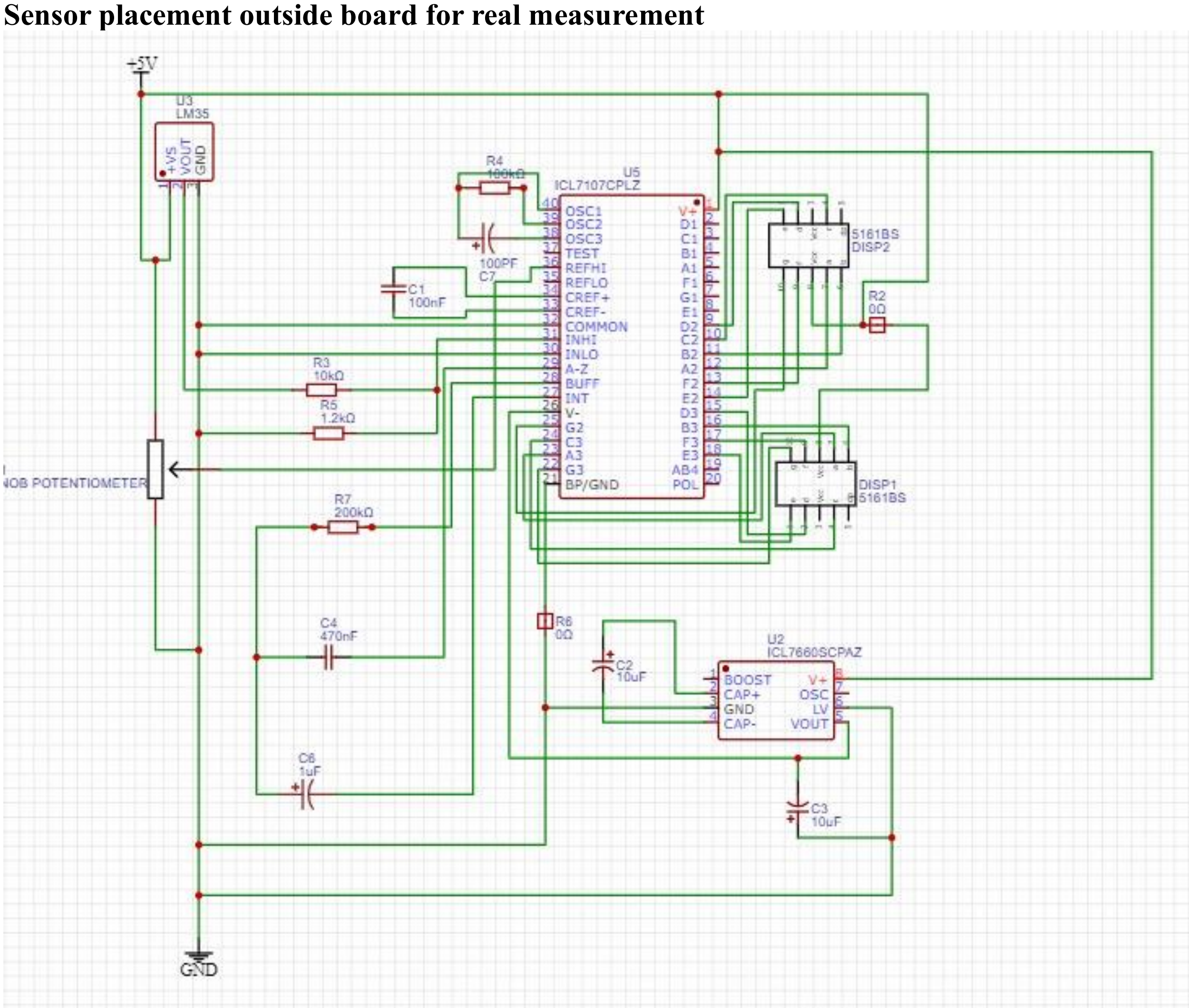
4.3 EasyEDA PCB Design

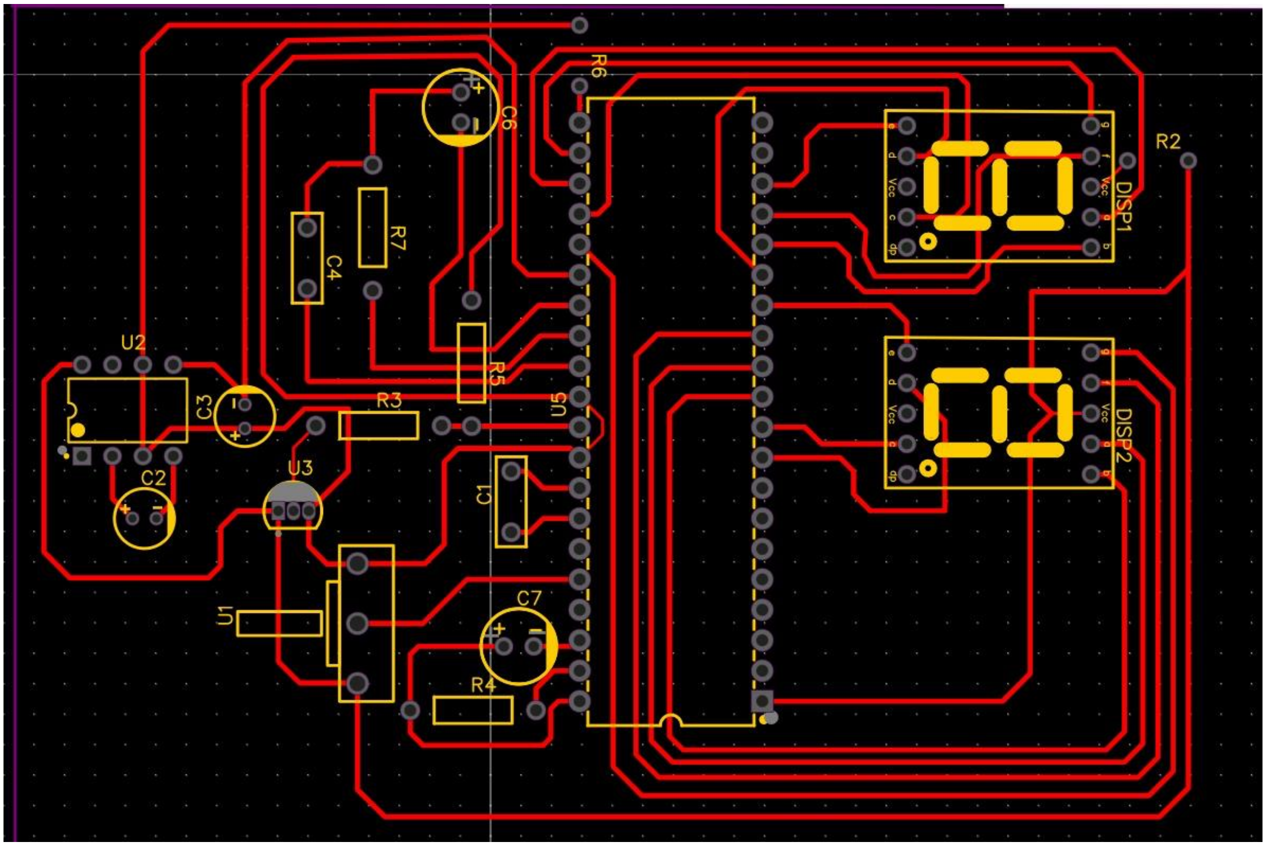
The circuit was converted into a PCB using EasyEDA Pro.

Key considerations:

- Proper component placement
- Short analog signal paths
- Clean routing

Special attention was given to:

- **Display alignment**
- **Sensor placement outside board for real measurement**
- 


R6
C6
R2
DISP1
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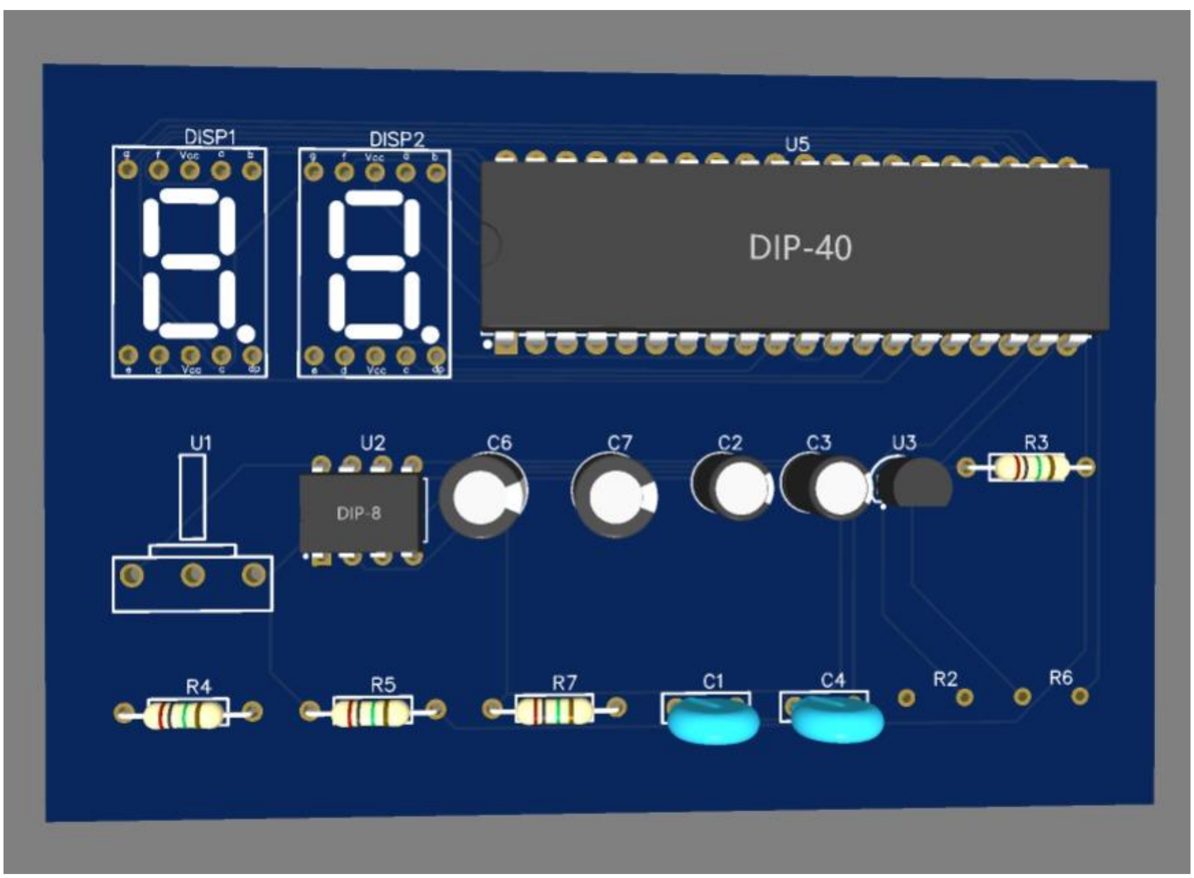
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DIP-40
U1
U2
C6
C7
C2
C3
U3
R3
DIP-8
R4
R5
R7
C1
C4
R2
R6

5. Component Selection

Component	Quantity	Package	Purpose
ICL7107	1	DIP-40	ADC + Display driver
LM35	1	TO-92	Temperature sensing
7-Segment Displays	2	Common Anode	Output display
Resistors/Capacitors	Multiple	SMD 0805	Filtering & stability
Potentiometer	1	3296W	Vref adjustment

Why we selected :

DIP IC → easier soldering

SMD → compact PCB

Pot → fine calibration control

6. Results & Testing

6.1 Simulation Results

The system showed accurate readings across full range:

0°C → 0 V → "00"

50°C → 500 mV → "50"

99°C → 990 mV → "99"

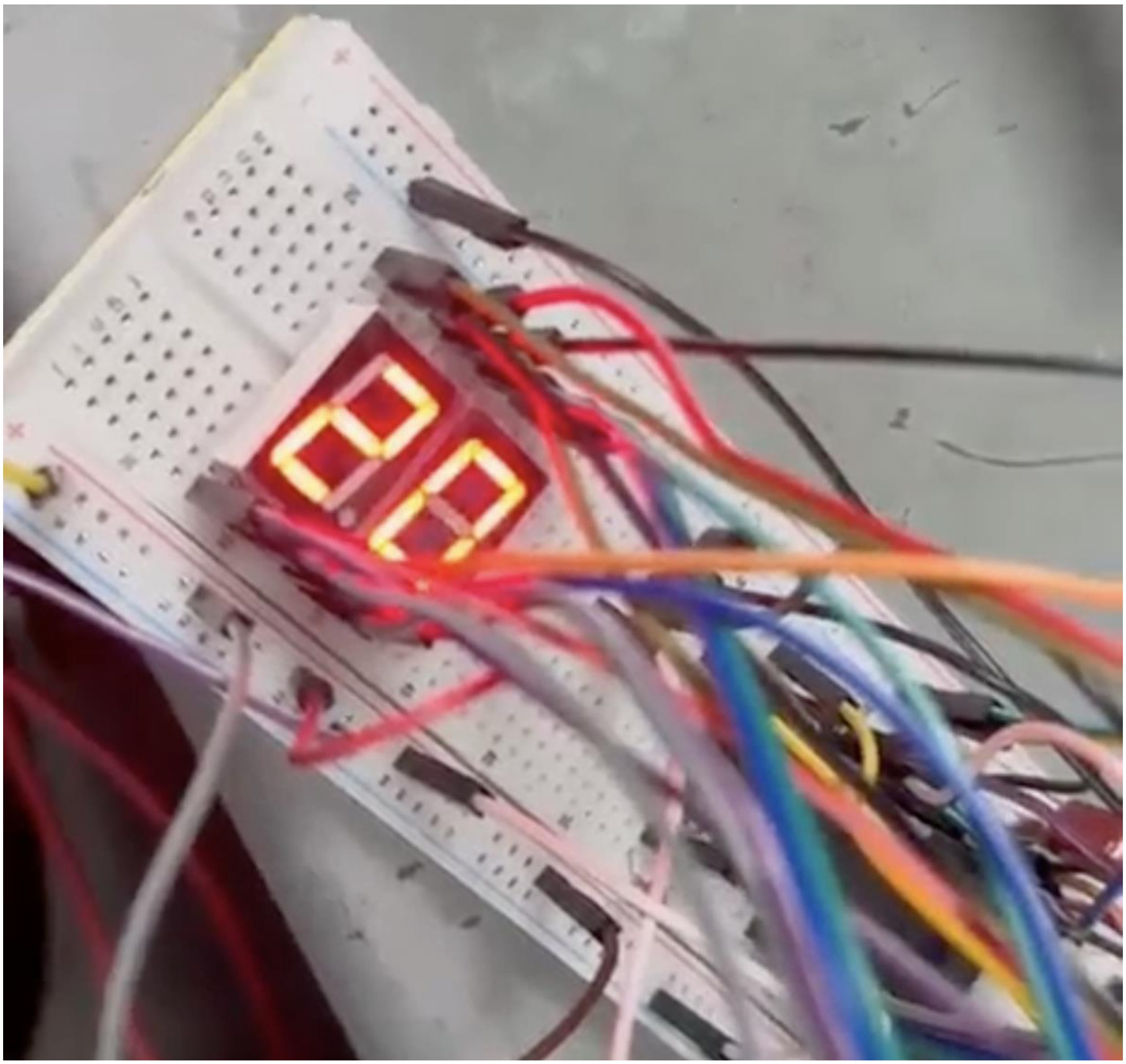
No fluctuation observed in simulation.

6.2 Actual Circuit Testing

The physical circuit was tested after PCB assembly.

Observations:

- Display responded correctly to temperature changes
- Minor fluctuations reduced after grounding improvement
- Calibration adjusted using potentiometer



7. Discussion

During the implementation of the digital thermometer, several practical issues were observed that were not visible in simulation. Although the Proteus model produced accurate and stable outputs, the real circuit showed sensitivity to small voltage variations. Since the LM35 operates at a very low scale of 10 mV per °C, even minor fluctuations in the power supply or external noise resulted in slight variations in the least significant digit of the display.

A common grounding method was used in this design. While the circuit functioned correctly, this approach may have allowed interference between the digital and analog sections. In mixed-signal systems, switching noise from the display driver can affect the low-level analog signal from the sensor. A more advanced grounding technique, such as star grounding, could reduce this effect and improve stability.

The requirement of a dual power supply (±5V) also introduced practical challenges. The negative supply is necessary for accurate readings near 0°C, but maintaining a stable negative voltage was difficult and influenced measurement accuracy.

Additionally, calibration of the reference voltage to exactly 100 mV required careful adjustment. Even small errors in Vref caused noticeable deviations in the output, highlighting the importance of precise calibration in analog systems.

8. Conclusion

The project successfully achieved its objective of developing a functional digital thermometer capable of measuring temperatures from 0°C to 99°C. Beyond simulation, the implementation highlighted key practical challenges such as noise sensitivity, grounding limitations, and calibration accuracy. These factors emphasized the importance of proper analog design techniques in real-world systems. Overall, the project provided valuable hands-on experience in integrating analog sensing with digital display systems and reinforced the gap between theoretical design and practical implementation

9. References

- LM35 Datasheet - (https://www.ti.com/lit/ds/symlink/lm35.pdf)

- ICL7107 Datasheet –(https://www.renesas.com/en/document/dst/icl7106-icl7107-icl7107s-datasheet?srsltid=AfmBOoqWwdulg1Psn1l813GUBhveu1nXSH4FmudgQ2ki1Uss4gPNMzhK)

- Lecture Notes

- Proteus & EasyEDA Documentation